JMT 2022 HeliosX

ENDING AND RECOVERING FROM MARINE DEBRIS

HeliosX

NVS Praseeth Sashank

Ramanathan AS

Nitya Kairavi V

Riddhima Prasad AS

The Problem

Increase In Marine Debris

Increase in marine debris:

- Adversely affects the marine ecosystem
- Disrupts economies and industries concerned with the seas and oceans
- Contributes to climate change
- Directly impacts human health via food chain
- Etc.

Our Solution

Agnosagua & Apokathisto

Advanced, eco-friendly, economic ocean rapid-cleaning buoy and segregation and sending of retrieved wastes to recyclers/decontaminators.

Steps involved in making the Prototype (Agnosagua)

COMPONENTS TO BE MADE:

- Tesla valve
- Fine Sieve
- Torque promoter
- Converging-diverging nozzle
- Electrostatic precipitator

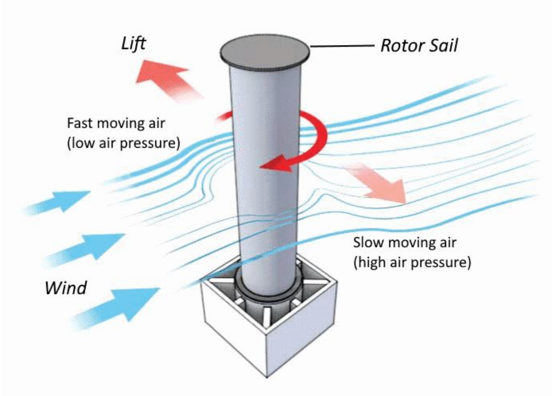
Assembly of Buoy

- Take 2 PVC pipes of diameter 3 inch
- Attach the two pipes using adhesive glue with a pipe connecting them
- Make apertures for the pipes and connect servos for opening and closing
- Make bottom pipe watertight
- Attach pipes to floats and attach 2 other floats , connected with hinges , micro servos for the storm mechanism
- Attach stirling engine with cassegrain apparatus and flettner rotor
- Connect the wheel of the stirling engine with bands to propellers at the bottom
- Connect and program the arduino, solar panels, load , tilt and acceleration sensors, GPS, etc.
- Fix in electrostatic precipitators

Parts of Prototype

S.No.	Component	Use
1	Stirling Engine	Powers the circuit board; Provides power to the propellers
2	Sails	Propels the machine
3	Circuit Board	Coordinates activities like the storm mechanism, sensors, sends capacity-full message
4	Solar Panels	Powers the circuit board
5	Set of Mirrors/ Lenses	Supply sufficient heat to stirling engine by concentrating heat
6	Flettner Rotor	Propels the machine
7	Thermos Insulation	Maintains constant temperature- colder than the upper plate of the Stirling engine, providing temperature differential; minimizes heat losses
8	Motor	Kick Starts the Stirling Engine, Draws in cables in the Storm Protection Mechanism, Turns sieves around when they are full

Stirling Engine with set of Mirrors



Flettner Rotor

S.No.	Component	Use
9	Greenhouse case	Using the greenhouse effect, heats up the top plate
10	Torque Promoter	Helps flywheel retain more torque
11	Primary Floats	keep the buoy afloat
12	Secondary Floats	part of storm mechanism- serve as shields against storms
13	Aperture	Closes the entrance and exit during storms
14	Coarse Sieve	Filters marine debris larger than 5mm, coarsely fiters the particles
15	Marine Life Exit	outlet marine life escape so that marne life is unaffected by the buoy
16	Filters	filter out wastes and protects the internal components to increase lifespan
17	Fine Sieve	filters microparticles

Floats

Fine sieve

Coarse Sieve

Stirling engine with torque promoter and solar panels

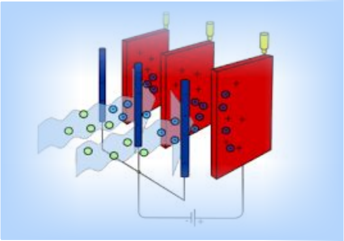
S.No.	Component	Use
18	Tesla Valve	prevents backflow of water containing wastes during night when temperature differential is reversed, during storms, and prevents wastes from reaching the propellers, reduces back-currents
19	Propellers	propel the buoy and suck in the wastes
20	Converging Diverging Nozzle	Increase velocity of water, help in contributing speed and efficiency, constrict the space for passage of microparticles
21	Tanks	2 tanks filter together acting as backups to each other, dump wastes in third tank that ac
22	Electrostatic Precipitator	nuclear, metallic, volatile, and some chemical wastes filtered
23	Ultrasound sensor	senses when coarse sieves in both tanks are full
24	Float sensor	senses when tank is full by detecting the “sinkage”
25	Hall & tilt sensors	detects when there is a storm

Tesla Valve

Float Sensor

Tilt Sensor

Ultrasound sensor



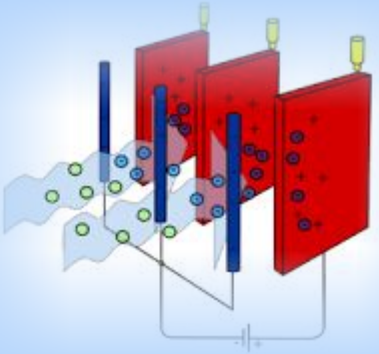
Electrostatic Precipitator

Tech used

Tech used

- Stirling Engine
- Flettner Rotor
- Tesla Valve
- Concentrated Solar Energy
- Greenhouse Effect
- Electrostatic Precipitator
- Biomimicry Of Baleen Structures Of Whales
- Torque Promoter
- Converging Diverging Nozzle
- Catamaran-type streamlining
- Cassegrain Apparatus

Baleen whale



How Uniquely It Solves the Problem

Unique Selling Proposition

- Revolutionary tech
- Efficient
- Eco-friendly
- Reduces landfills
- Cheap
- Large working radius
- Minimal energy wastage
- Faster cleanup in larger quantities
- Microplastic, chemical waste, and nuclear waste cleanup
- Active-passive hybrid, semi-autonomous

Cost Breakdown

Pipes-200

DIY parts-1000

float sensor -100

carpented parts-2000

Stirling engine-2700

Robotic parts:-

Solar panels(2)-450

Micro servos(2)-260

Load cell-200

float sensor-100

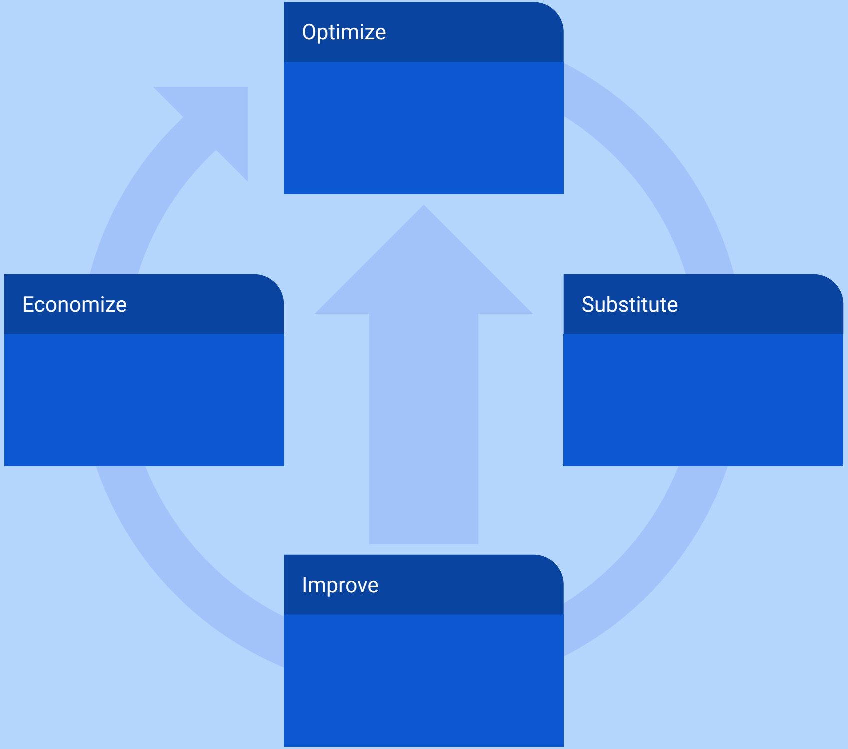
Arduino Mega-970

Total-8590 INR*

*Costs are far decreased when scrapyard material is used instead, as for the industry-grade machine

Changes Needed to Scale-up

Changes needed to scale-up



Customers

Customers

B2B

Ecologically conscious Organizations, Reusing and Recycling Organizations, Decontaminators

B2C

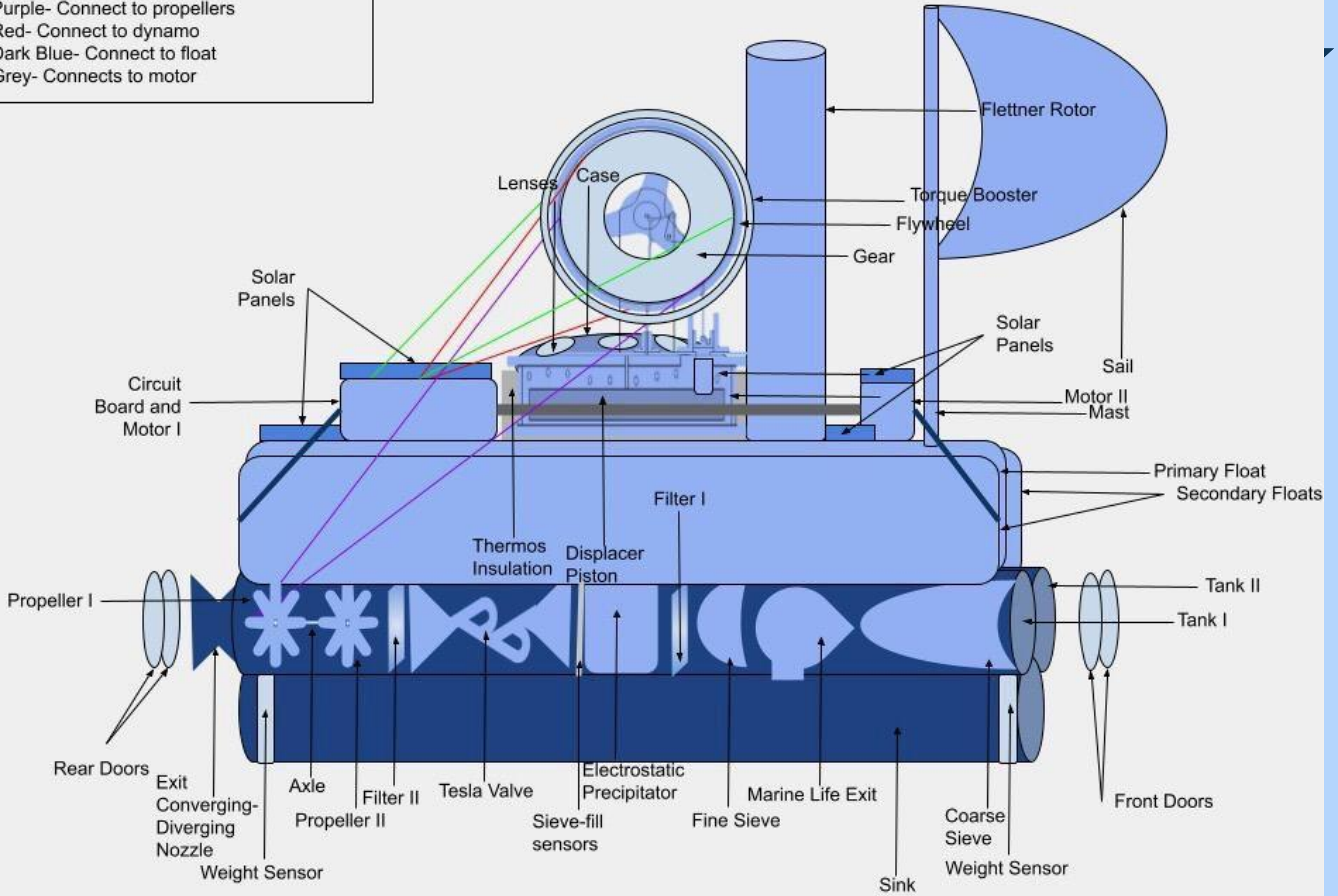
Individual Marine Life Activists, Volunteers, Recyclers, Retailers

Scope


North Pacific
Subtropical Convergence Zone
Kuroshio
Western Garbage Patch
California
Eastern Garbage Patch or
N. Pacific Subtropical High
North Equatorial

Legend for Colouring of Lines

Green- Connect to motor
Purple- Connect to propellers
Red- Connect to dynamo
Dark Blue- Connect to float
Grey- Connects to motor



Thankyou!

“The sea cures all ailments of man”

- Plato